MIL-E-1/970D
27 December 1976
SUPERSEDING
MIL-E-1/970C
16 April 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 1B58A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: TR, bandpass, frequency range 2,664 to 2,964 MHz, incident power 750 kW.

ABSOLUTE RATINGS:

Parameter:	Incident power	Ebb (open circuit)	Alt
Unit:	kW	Vdc	ft
Maximum:	--	---	10,000
Minimum:	10	-700	---

PHYSICAL CHARACTERISTICS: See figure 1

TEST CONDITIONS:

Parameter:	F	tp1	tp2	prr
Unit:	MHz	µs	µs	pps
Test condition 1:	F4 ±0.5%	1.0 ±0.15	0.5 ±0.15	1,000
Test condition 2:	F4 ±0.5%	1.0	---	1,000
Test condition 3:	F4 ±0.5%	---	---	---

Parameter:	Du	Incident power	Ii	Ri
Unit:	----	kW	µAdc	meg
Test condition 1:	----	200 ±10	200	1.6
Test condition 2:	0.001	50	200	1.6
Test condition 3:	0.0008	750 ±10%	200	1.6

GENERAL:

Qualification - Required

TEST FREQUENCIES	
F	MHz tolerance ±0.1%
1	2664
2	2689
3	2714
4	2802
5	2914
6	2939
7	2964

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	METHOD	REQUIREMENT OR TEST	TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
		Qualification inspection						
	4021	Degradation due to vibration	--		---	---	---	---
	4774	High-level VSWR	2	Note 2	---	---	1.15	---
		Quality conformance inspection, part 1	--	Note 5				
	4401	Ignitor ignition time	--	Ebb = -600 Vdc;	t	---	5.0	sec
D	4406	Ignitor voltage drop	--	Ii = 200 µAdc	Eid	200	400	Vdc
	4452	Spike-leakage energy	1	Note 2	Ws	---	0.3	erg
	4452	Flat-leakage power	1	Note 2	pf	---	40	mw
	1027	Temperature cycling (nonoperating)	--	1 cycle	---	---	---	---
D	4473	Low-level VSWR	--	F1 F2 to F3 F3 to F5. F5 to F6 F7 Notes 1 and 3	--- --- --- --- ---	--- --- --- --- ---	1.65 1.2 1.3 1.2 1.65	--- --- --- --- ---
		Quality conformance inspection, part 2						
	4101	Dielectric material strain	--		---	---	---	---
	4003	Pressurizing	--	50 lb_f/in^2, Note 4	---	---	---	---
D	4416	Insertion loss (fixed tuned)	--	F4; Ii = 0 Note 1	Li	---	0.3	dB
D	4421	Ignitor interaction (insertion loss)	--	Ii = 200 µAdc	Li	---	0.1	dB
	4471	Recovery time	3	Ii = 200 µAdc Note 2	t	---	15	µs
	----	Insertion	--	Note 6	---	---	---	---
		Quality conformance inspection, part 3						
D	----	Life test	3	Group D; Ii = 150 to 200 µAdc; Note 2	t	1,000	---	hrs
	----	Life-test end points						
D	4416	Insertion loss (fixed tuned)	--	F4; Ii = 0 Note 1	Li	---	1.0	dB
	4452	Spike-leakage energy	1	Note 2	Ws	---	0.3	erg
	4452	Flat-leakage power	1	Note 2	pf	---	40	mw
	4471	Recovery time	3	Note 2	t	---	30	µs
	1027	Temperature cycling life-test end point	--	Group C; 10 cycles	---	---	---	---

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NOTES:

1. This test shall be performed using the flanges specified in Drawing 268-JAN.
2. This test shall be performed using the mount specified in Drawing 153-JAN.
3. A swept frequency method of measurement may be used instead of measurement at fixed frequencies.
4. The tube shall be mounted per Drawing 208-JAN, and the complete assembly cycled once from -55° to +100°C. After the temperature has again reached room temperature, the pressure shall not have changed more than one-quarter of a pound as indicated by a gage permanently connected into the system.
5. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.
6. The tube shall be capable of being inserted and removed from the mount specified in Drawing 153-JAN a minimum of 15 times with no deterioration in the tube's electrical characteristics.

Custodians:

Army - EL
Navy - EC
Air Force - 85

Preparing activity:

Air Force - 85

Review activities:

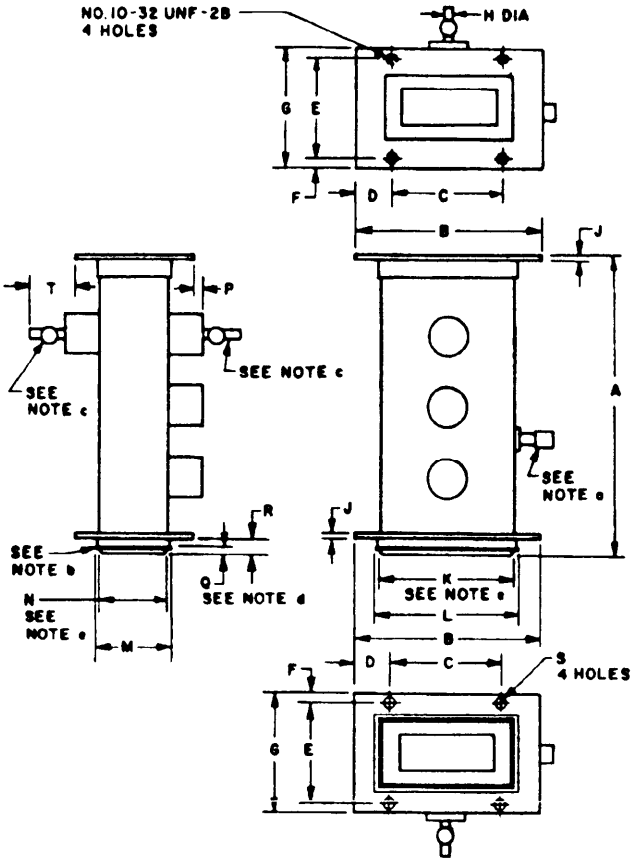
Army - MU
Air Force - 99
DSA - ES

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User activities:

Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

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LTR	DIMENSIONS IN INCHES WITH METRIC EQUIVALENTS (mm) IN PARENTHESES	
	MINIMUM	MAXIMUM
QUALITY CONFORMANCE INSPECTION, PART 1 (NOTE 5)		
A	6.580 (167.13)	6.640 (168.66)
K		3.008 (76.40)
P		.19 (4.83)
Q	.165 (4.19)	.195 (4.95)
R	.347 (8.81)	.377 (9.58)
T		.81 (20.57)
QUALITY CONFORMANCE INSPECTION, PART 2		
C	2.490 (63.25)	2.510 (63.75)
E	2.240 (56.90)	2.260 (57.40)
L	3.203 (81.36)	3.233 (82.12)
M	1.703 (43.26)	1.733 (44.02)
N		1.508 (38.30)
REFERENCE DIMENSIONS		
B	4.12 (104.65)	
D	.81 (20.57)	
F	.19 (4.83)	
G	2.62 (66.55)	
H	.25 (6.35)	
J	.12 (3.06)	
S	.22 (5.59)	

NOTES:

a. Exhaust tube shall not extend beyond flange more than .25 (6.35 mm) inch.
b. Gasket per Drawing 189-JAN. Gasket to be securely attached.
c. Ignitor electrode may be mounted in either of positions shown.
d. Dimensions Q shall be measured prior to the attachment of the gasket to the tube.
e. Edges may be rounded.
f. Mount either series or shunt mount may be used. If series mount is used, mount shall be in accordance with Drawing 153-JAN.

FIGURE 1. Outline drawing of electron tube type 1B58A.

*1

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-970D Electron Tube, Gas Switching Type 1B58A FSC 5960

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